Freight Rate Prediction Challenge - Technical Report

1. Problem Framing

Predict 12,000 freight loads (Nov–Dec 2025) + 31-day Dec chart for Lexington→Fort Wayne (360 mi, Dry Van, 32,000 lbs). Training: 48,000 loads (Jan–Oct 2025). Metric unspecified; MAPE assumed.

2. Data & EDA

Train: 48k, Jan–Oct 2025. Val: 12k, Nov–Dec 2025. Dec chart: 31 rows, fixed lane.

2.1 Key Findings

Distance dominates: Corr 0.91. Rate ≈ distance × RPM + . RPM: Dry Van 2.12, Reefer 2.38, Flatbed 2.29
Market Index day-level: Within-day std 0.025. Daily MI + trend explains 59% daily RPM variance. Row-level corr ~0.03
Seasonality: RPM 2.10 Jan → 2.33 Jun → 2.24 Oct. No Nov/Dec history
Distribution shift: Val MI 0.93 vs Train 1.08 (14% cooler). QS range truncated
Cold start: 8 new cities, 17.5% unseen lanes. Geo + smoothed encodings handle this
Heavy tail: Rate –,533, mean ,374. Log target used
Dec chart lane: Lex>FW: 32 rows, –. No Dec history

3. Validation

Strict temporal. Expanding-window CV: Jan–Jun|Jul, Jan–Jul|Aug, Jan–Aug|Sep, Jan–Sep|Oct. OOF encodings. No leakage verified. Stress test: warm→cool MAPE 10.1% vs 6.1% normal.

4. Features

Distance, geo, equipment, weight (abs, clip, short-flag), temporal (cyclical), market (daily MI/QS), OOF target encodings, freq encodings, lane/city stats, holiday-distance (dampened for short). Target: log(rate).

5. Results

Model	MAPE (%)	MAE ($)	RMSE ($)	MedAPE (%)	R^2
LightGBM	6.06 ± 0.47	137 ± 7	636 ± 11	2.99 ± 0.41	0.822 ± 0.006
XGBoost	6.37 ± 0.47	144 ± 9	639 ± 13	3.17 ± 0.30	0.820 ± 0.007
Ridge (numeric)	19.7 ± 0.9	434 ± 15	942 ± 53	12.2 ± 2.4	0.609 ± 0.046
Ensemble (inv MAPE)	6.86 ± 0.48	149 ± 13	640 ± 13	3.60 ± 0.30	0.820 ± 0.007

LightGBM best single (6.06% MAPE). Ensemble (6.86%) worse; final: LightGBM only, 5-seed avg.

6. Error Analysis

• MAPE by month: lowest Sep 5.5%, highest Jul 6.8%
• Unseen lanes (17.5%): within 10% of seen
• Short dist (<150 mi): no neg preds (log target)
• Top SHAP: distance, market_index, haversine, month_sin, equipment

7. December Chart

Predicted Dec: – (mean) vs historical – (mean). Captures DOW. No Dec history. Smooth curve.

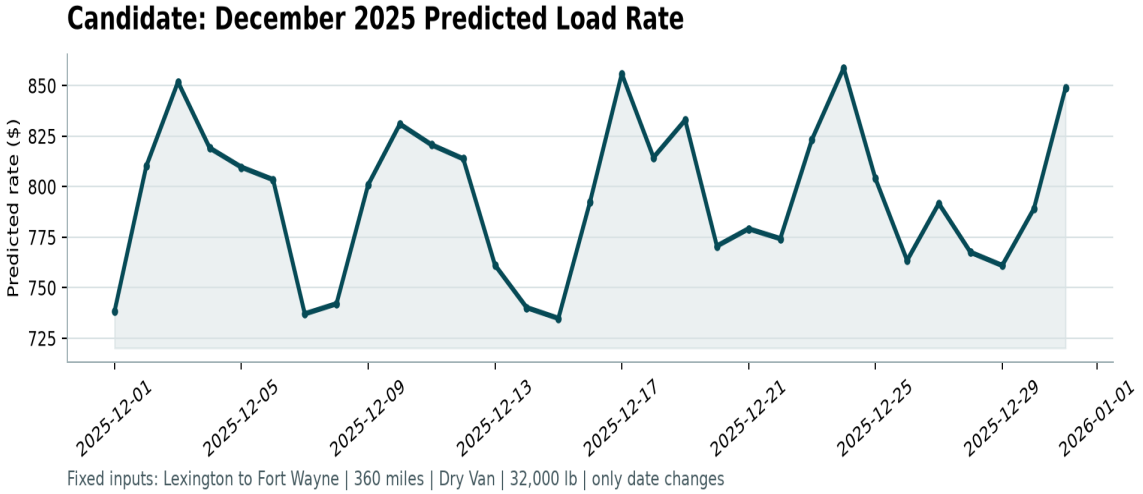


Figure 1: Candidate December 2025 Predicted Load Rate

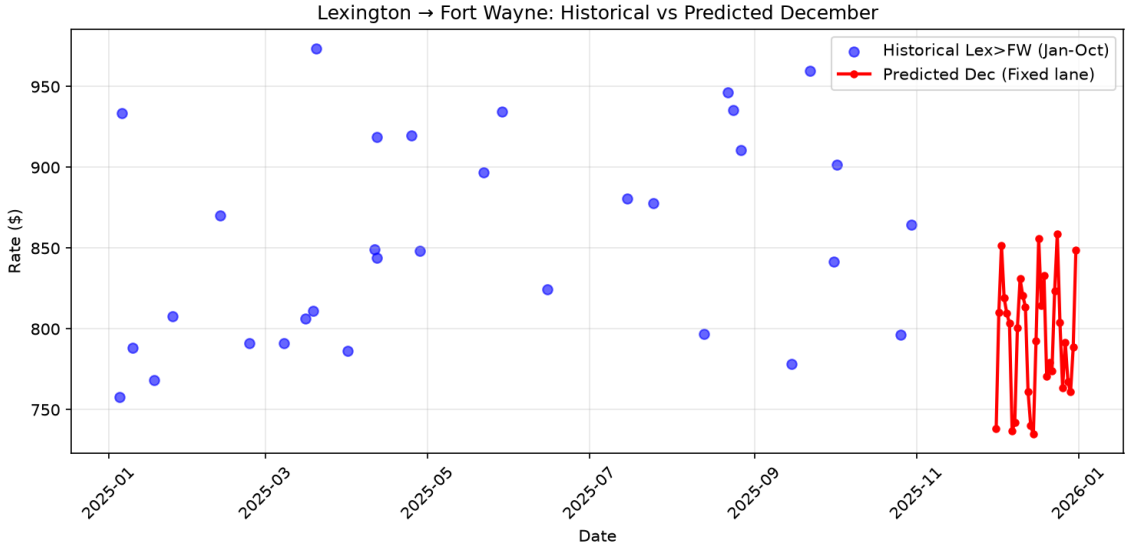


Figure 2: Historical vs Predicted December

7.1 Missing Coordinates

Dec chart lacks lat/lon. Look up from training (each city has consistent coords). Lexington/Fort Wayne in training → exact recovery. Static features, no leakage.

7.2 Jaggedness Fix

Initial spike Dec 17 (~,085), + swings. Cause: day_of_year (train 1–304) and week (1–44) used, but Dec has 335–365 and 49–52 — pure extrapolation. Fix: replaced with dow + holiday-distance. Reg: num_leaves=255, min_child_samples=50, subsample=0.8. 5-seed avg. Result: smooth –.

8. Limitations

• No Dec history: extrapolation
• MI shift 0.93 vs 1.08
• LightGBM only vs ensemble
• Feature store not productionized
• Limited HPO
• Weight sign-fix assumption (negative = typos)

9. Next Steps

• Collect Nov/Dec data
• Optuna tuning
• Add fuel/weather/holiday
• Feature store
• Quantile regression
• A/B test ensemble